Somaiya Vidyavihar University
K J Somaiya School of Engineering

FY B Tech (Common to All)
SVU R-2025 Version 3.0
SEM II

Teaching and Credit Scheme

Course Code		Course Category	Name of the Course	Teaching Scheme TH-PR-TUT	Total (hrs.)	Credit Scheme TH-PR-TUT	Total Credits
SVU	KJSSE						
16U1012	316U06C201	BS	Applied Mathematics – II	3 – 0 – 1	4	3 – 0 – 1	4
16U1022 16U1023 16U1018	316U06E211 316U06E212 316U06E221	BS BS ES	Applied Science for Computer & Allied Programs/ Applied Science for Electronics & Allied Programs/ Engineering Mechanics †	3 – 0 – 0	3	3 – 0 – 0	3
16U1019	316U06C203	ES	Digital Logic Design	3 – 0 – 0	3	3 – 0 – 0	3
16U1013	316U06C204	HSS	Environmental Science	2 – 0 – 0	2	2 – 0 – 0	2
16U1014	316U06C205	ES	Object-Oriented Programming Methodology	2 – 2 – 0	4	2 – 1 – 0	3
16U1015	316U06T201	HSS	Presentation and Communication Skills	1 – 0 – 1	2	1 – 0 – 1	2
16U1016 16U1021	316U06L201 316U06L221	BS ES	Applied Science Laboratory - II/ Engineering Mechanics Laboratory	0 – 2 – 0	2	0 – 1 – 0	1
16U1020	316U06L202	ES	Digital Logic Design Laboratory	0 – 2 – 0	2	0 – 1 – 0	1
16U1017	316U06L203	ES	Maker Space Laboratory – II	0 – 2 – 0	2+1#	0 – 1 – 0	1
	--		SVU Basket Course				
			Total	14 – 8 – 2	24+1#	14 – 4 – 2	20

† Only for MECH/RAI students (in place of Applied Science)

#Additional 1 hr/division/week is assigned for project review of PBL component (only for timetable slot. No separate course credit is given)

Examination Scheme

Course Code		Course Category	Name of the Course	Lab/ Tut CA	CA		ESE		Total
SVU	KJSSE				IA	MSE	PR/OR Exam/ OST/ Project	Theory Exam	
16U1012	316U06C201	BS	Applied Mathematics – II	25	20	30	--	50	125
16U1022 16U1023 16U1018	316U06E211 316U06E212 316U06E221	BS BS ES	Applied Science for Computer & Allied Programs/ Applied Science for Electronics & Allied Programs/ Engineering Mechanics †	--	20	30	--	50	100
16U1019	316U06C203	ES	Digital Logic Design	--	20	30	--	50	100
16U1013	316U06C204	HSS	Environmental Science	--	50	--	--	--	50
16U1014	316U06C205	ES	Object Oriented Programming Methodology	50	--	--	50	--	100
16U1015	316U06T201	HSS	Presentation and Communication Skills	50	--	--	--	--	50
16U1016 16U1021	316U06L201 316U06L221	BS ES	Applied Science Laboratory - II/ Engineering Mechanics Laboratory	50	--	--	--	--	50
16U1020	316U06L202	ES	Digital Logic Design Laboratory	50	--	--	--	--	50
16U1017	316U06L203	ES	Maker Space Laboratory – II	50	--	--	50$	--	100
	—		SVU Basket Course						
			Total	275	110	90	100	150	725

$Project evaluation applicable only for PBL component of Makerspace Laboratory

<table>
<tr><th>Course Code</th><th colspan="6">Course Title</th></tr>
<tr><td>316U06C204</td><td colspan="6">Object-Oriented Programming Methodology</td></tr>
<tr><td rowspan="2">Teaching Scheme (hrs.)</td><td>TH</td><td colspan="2">PRACT</td><td colspan="2">TUT</td><td>Total</td></tr>
<tr><td>02</td><td colspan="2">02</td><td colspan="2">--</td><td>04</td></tr>
<tr><td>Credits Assigned</td><td>02</td><td colspan="2">01</td><td colspan="2">--</td><td>03</td></tr>
<tr><td rowspan="4">Examination Scheme</td><td colspan="6">Marks</td></tr>
<tr><td colspan="2">CA</td><td rowspan="2">ESE (On-Screen*)</td><td rowspan="2">PR/OR Exam</td><td rowspan="2">LAB/TUT CA</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>MSE</td></tr>
<tr><td>--</td><td>--</td><td>50</td><td>--</td><td>50</td><td>100</td></tr>
</table>

*The End Semester Examination (ESE) will be conducted in an onscreen format and may include a combination of theoretical questions, practical coding tasks, and/or an oral/viva component

Course prerequisites:

Students should have basic knowledge of:

- Programming fundamentals
- Data types, loops, control structures
- Basic understanding of algorithms and flowcharts

Course Objectives:

- Understand the principles and paradigms of object-oriented programming.
- Learn the structure and relationships of classes, objects, and methods.
- Apply concepts of inheritance, polymorphism, abstraction, and encapsulation in software development.
- Explore exception handling, multithreading, and file management in an object-oriented environment.

Course Outcomes

At the end of the successful completion of the course, the student will be able to

CO1. Understand the fundamentals of object-oriented programming and complex system design.
CO2. Identify and apply classes, objects, methods, and constructors to develop modular programs.
CO3. Implement advanced OOP concepts such as inheritance, polymorphism, and abstract classes.
CO4. Implement applications involving multithreading, exception handling, file operations, and reusable components.

Module No.	Unit No.	Contents	Hrs.	CO
1	**Fundamentals of Object-Oriented Programming**		05	CO1
	1.1	Introduction, Basic Program Construction, Procedural Programming Approach, Structured Programming Approach, Modular Programming Approach, OOP Approach		
	1.2	The Structure of Complex Systems with attributes, The Inherent Complexity of Software		
2	**Classes and Objects, methods and constructors**		08	CO2
	2.1	The Object Model, Elements of the Object Model – (Features) Applying the Object Model		
	2.2	The Nature of an Object, Relationships among Objects, The Nature of a Class Relationships among Classes The Interplay of Classes and Objects		
	2.3	Identifying Classes and Objects Key Abstractions and Mechanisms		
	2.4	Member, method, State of an object. Memory allocation of an object		
3	**Inheritance and Polymorphism**		07	CO3
	3.1	Inheritance, Types of Inheritance		
	3.2	Method overloading & overriding, constructor, Types of constructors, Constructor Overloading, Concept of memory management of unused objects/references (destructor /Garbage collection), Early vs. Late Binding, Runtime polymorphism		
	#	**Self-Learning topic**: Template /Generic classes		
4	**Packages/namespaces, Exception Handling, Multithreading, files**		10	CO4
	4.1	Introduction to Packages/Namespaces.		
	4.2	Introduction to Exception Handling.		
	4.3	Multithreading: Thread life cycle, Multithreading advantages and issues, Simple thread program, Thread synchronization.		
	#	**Self-Learning topic**: The Standard Template Library, Various packages/ namespaces, File handling		
		Total	30	--

Note: Each theory topic should be supported by relevant programming examples delivering.

Recommended Books:

Sr. No	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1	Grady Booch, James Rumbaugh, Ivar Jacobson	*Unified Modeling Language*	Pearson Education	3rd Edition, 2007
2	Herbert Schildt	*Java The Complete Reference*	Tata McGraw-Hill	12th Edition, 2022
3	E Balagurusamy	*Object Oriented Programming in C++*	Tata McGraw Hill	5th Edition, 2011
4	Sheetal Taneja and Naveen Kumar	Python Programing: A Modular Approach	Pearson India	2nd Edition, 2018, India
5	Sachin Malhotra, Saurabh Chaudhary	Programming in Java	Oxford University, India	2nd Edition, 2018
6	NPTEL: C++	https://onlinecourses.nptel.ac.in/noc21_cs02/preview		
7	NPTEL: Java	https://onlinecourses.nptel.ac.in/noc22_cs47/preview		
8	NPTEL: Python	https://nptel.ac.in/courses/106106145		

Note:
1. The total marks assigned for Continuous Assessment (LAB/TUT) as per scheme can be distributed in a number of components as given above
2. ***All components should be same across all divisions.***
3. The course coordinator should decide the rubrics/distribution of marks for different components in consultation with all other faculty teaching the same course.
4. ***The course department should decide the language of implementation as per the requirement of the programme.***
5. The rubrics/distribution should be uniform for all batches.
6. The rubrics/distribution should be communicated to all students at the beginning of the semester.
7. C++/Java/Python is recommended to teach
8. No IA, ISE, ESE on self-learning topics, but self-learning topics can be covered in laboratory experiments.